MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

RESISTOR, ANALOG, KEY CODE GRB

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic resistor module.

1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

* 2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:42Z
Check the source to verify that this is the current version before use.

3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.020 failures per million operating hours for class I or 0.025 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be GRB.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/318-1 for class I and M28787/318-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

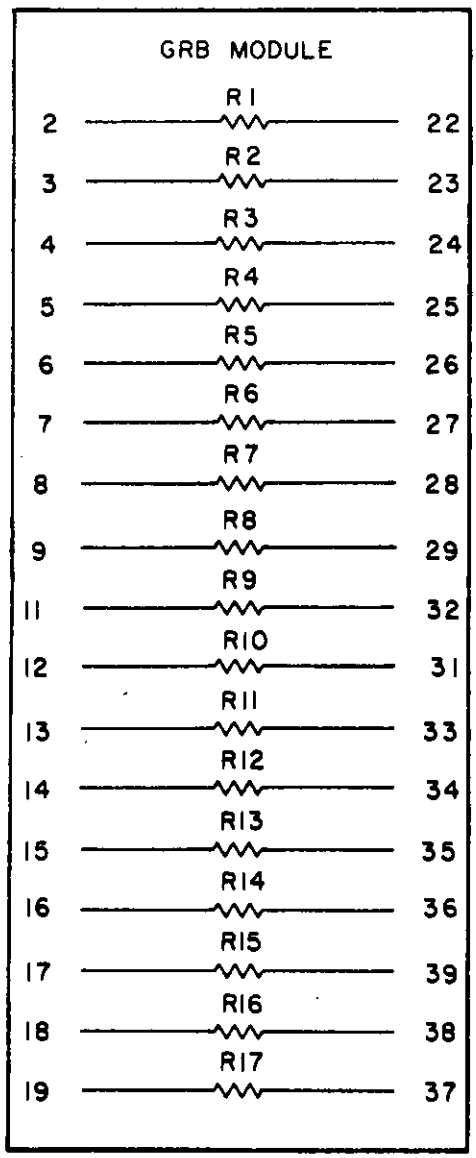
4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:42Z
Check the source to verify that this is the current version before use.



NOTES:
1. Unused pins: 10, 20, 21, 30, and 40.
2. R1-R17 = 1.0 kΩ ±1%, 1/10 W.

FIGURE 1. GRB module function diagram, resistor.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:42Z
Check the source to verify that this is the current version before use.

* TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/	25°±5°C		Class I				Class II				Unit
						0°±5°C		60°±5°C		-55°±5°C		85° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A1 - A18	Isolation													
	Initial	II	X	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	II	X	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B1-B17	Resistance													
	Initial	III	X	990	1,010	989	1,011	988	1,012	986	1,014	987	1,013	Ω
	End-of-life	III	X	985	1,015	984	1,016	983	1,017	981	1,019	981	1,019	Ω

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

Source: https://assist.dla.mil - Downloaded: 2026-10-20T17:42Z
Check the source to verify that this is the current version before use.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A and B shall be performed in any order with the following exceptions. If test condition A is part of a sequence of tests, the order of testing shall be as follows: A where applicable, followed by the remaining test in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A and B in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test condition B shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 2. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical test B shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests A and B shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

* 6.4 Module description. The following electrical parameters reflect the 0° to 60°C and -55° to 85°C end-of-life limits from table I.

	Class I	Class II	Unit
Maximum power dissipation	1.7	1.7	W
Maximum power dissipation per resistor (17)	0.1	0.1	W

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:42Z
Check the source to verify that this is the current version before use.

6.5 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5963-N365)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:42Z
Check the source to verify that this is the current version before use.

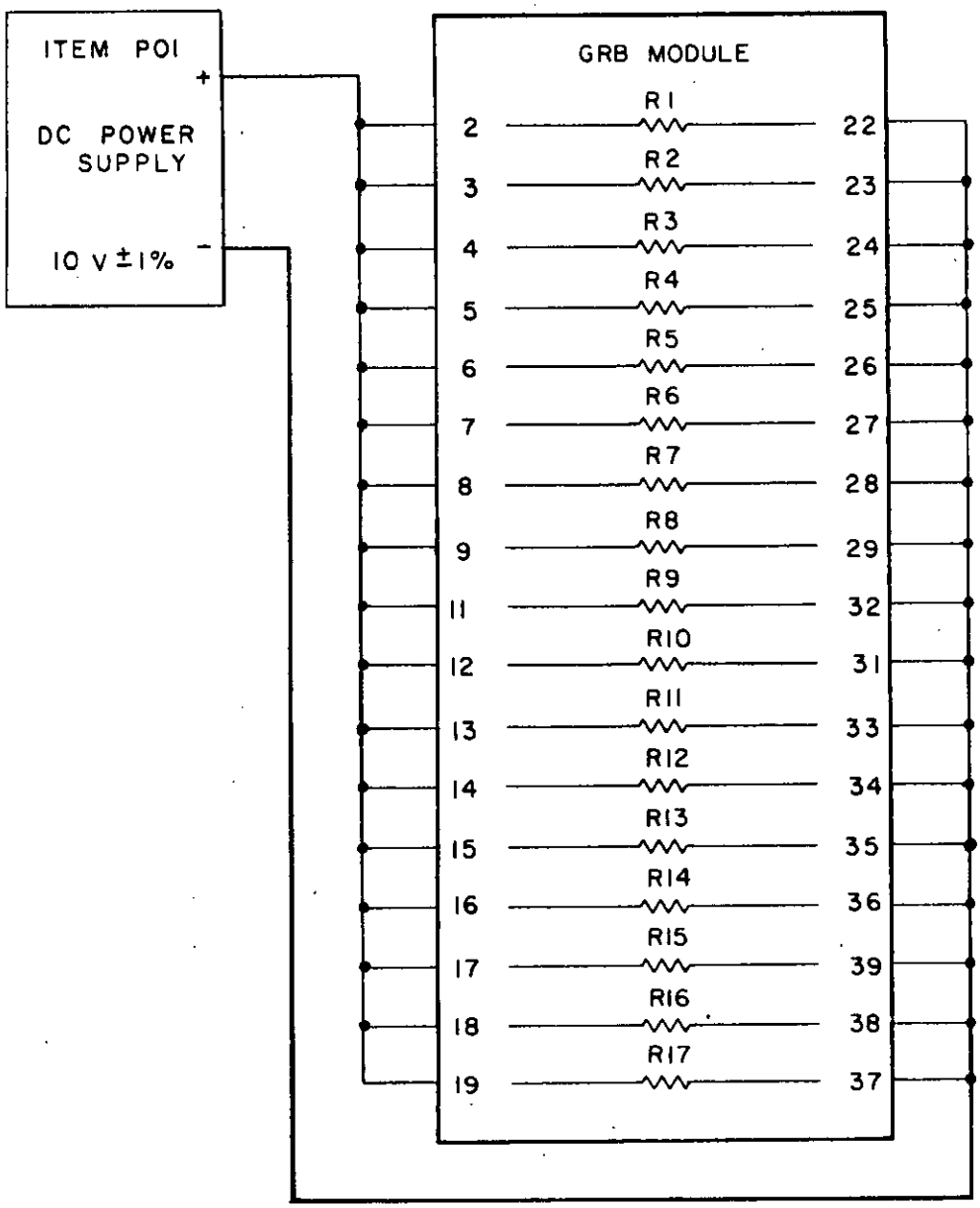


FIGURE 2. Operating temperature, component temperature, vibration and life test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:42Z
Check the source to verify that this is the current version before use.

TABLE II. Isolation test.

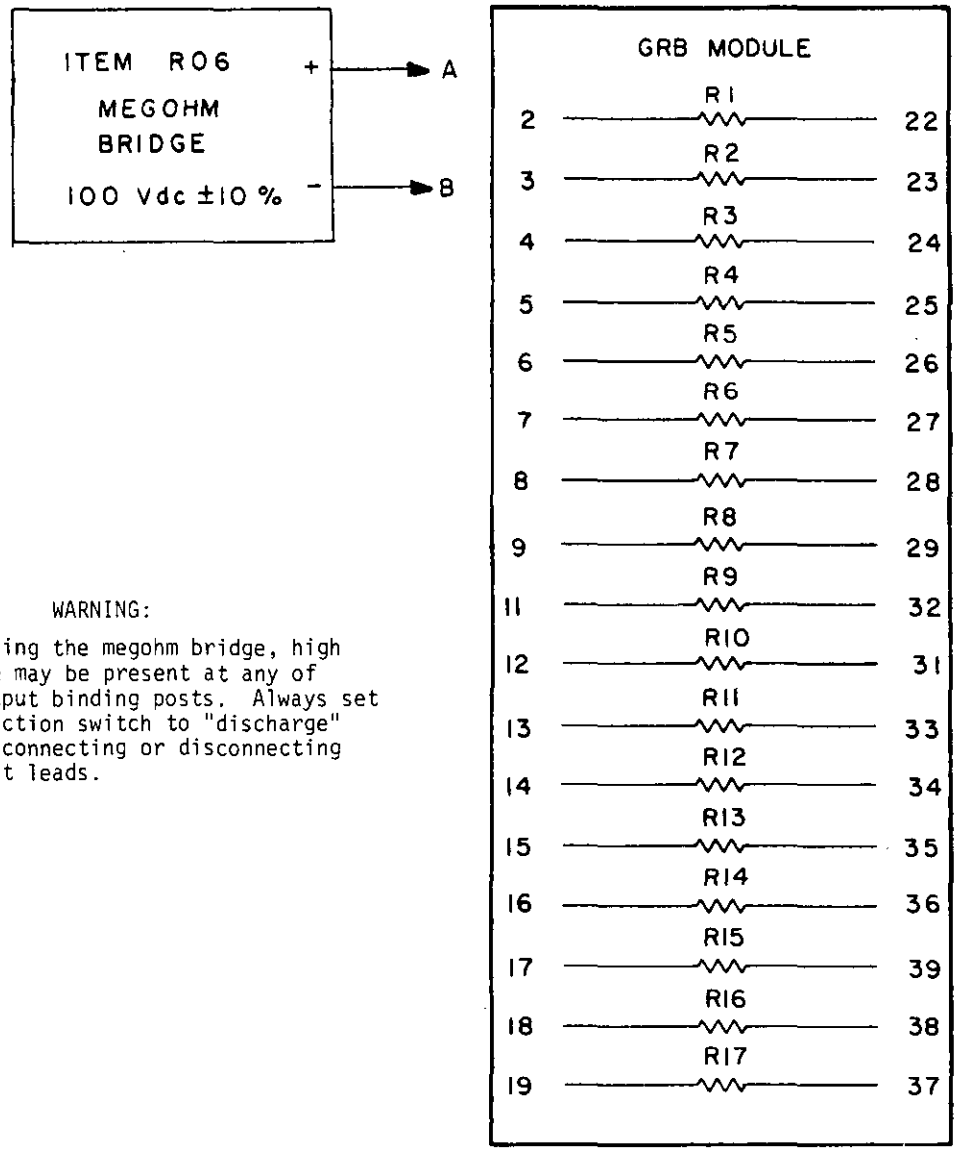
Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	R1 to other circuits Isolation	2, 22	3,4,5,6,7,8,9,11,12,13,14,15,16,17,18, 19,23,24,25,26,27,28,29,31,32,33,34,35, 36,37,38,39	10	∞	MΩ	3
A2	R2 to other circuits	3, 23	4,5,6,7,8,9,11,12,13,14,15,16,17,18,19, 24,25,26,27,28,29,31,32,33,34,35,36,37, 38,39				
A3	R3 to other circuits	4, 24	5,6,7,8,9,11,12,13,14,15,16,17,18,19,25, 26,27,28,29,31,32,33,34,35,36,37,38,39				
A4	R4 to other circuits	5, 25	6,7,8,9,11,12,13,14,15,16,17,18,19,26,27, 28,29,31,32,33,34,35,36,37,38,39				
A5	R5 to other circuits	6, 26	7,8,9,11,12,13,14,15,16,17,18,19,27,28, 29,31,32,33,34,35,36,37,38,39				
A6	R6 to other circuits	7, 27	8,9,11,12,13,14,15,16,17,18,19,28,29,31, 32,33,34,35,36,37,38,39				
A7	R7 to other circuits	8, 28	9,11,12,13,14,15,16,17,18,19,29,31,32,33, 34,35,36,37,38,39				
A8	R8 to other circuits	9, 29	11,12,13,14,15,16,17,18,19,31,32,33,34, 35,36,37,38,39				
A9	R9 to other circuits	11, 32	12,13,14,15,16,17,18,19,31,33,34,35,36,37, 38,39				
A10	R10 to other circuits	12, 31	13,14,15,16,17,18,19,33,34,35,36,37, 38,39				
A11	R11 to other circuits	13, 33	14,15,16,17,18,19,34,35,36,37,38,39				
A12	R12 to other circuits	14, 34	15,16,17,18,19,35,36,37,38,39				
A13	R13 to other circuits	15, 35	16,17,18,19,36,37,38,39				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:42Z
Check the source to verify that this is the current version before use.

TABLE II. Isolation test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A14	R14 to other circuits Isolation	16, 36	17,18,19,37,38,39	10	∞	MΩ	3
A15	R15 to other circuits	17, 39	18,19,37,38				
A16	R16 to other circuits	18, 38	19, 37				
A17	Used pins to unused pins	1,10,20, 21,30,40	2,3,4,5,6,7,8,9,11,12,13,14,15,16,17, 18,19,22,23,24,25,26,27,28,29,31,32, 33,34,35,36,37,38,39				
A18	Frame	All pins	Frame				

Source: https://assist.dla.mil - Downloaded: 2026-10-20T17:42Z
Check the source to verify that this is the current version before use.



WARNING:

When using the megohm bridge, high voltage may be present at any of the output binding posts. Always set the function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: For test condition A18, penetrate the anodized finish on the frame with a sharply pointed probe connected to the negative terminal.

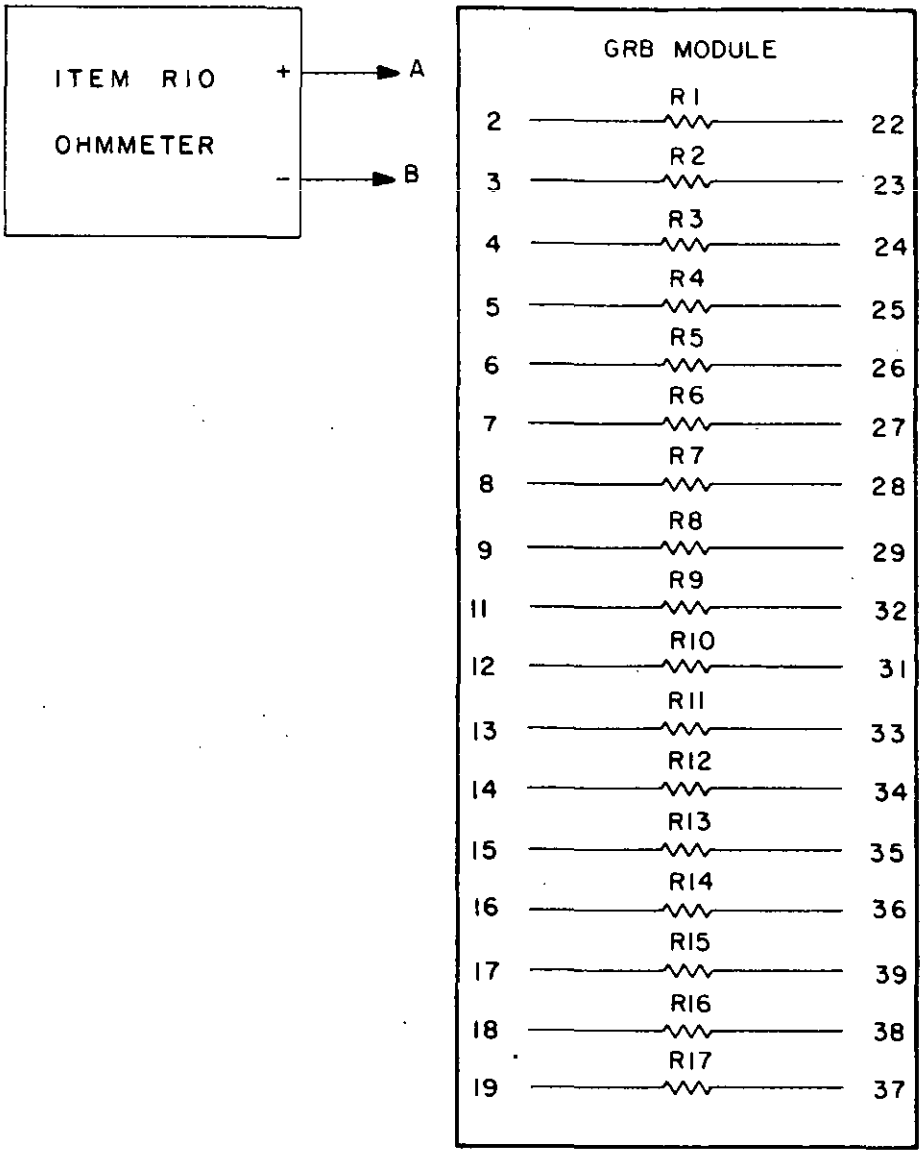
FIGURE 3. Isolation test circuit for table II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:42Z
Check the source to verify that this is the current version before use.

TABLE III. Resistance test.

Test condition	Parameter		Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
B1	R1	Resistance	2	22	990	1,010	Ω	4
B2	R2		3	23				
B3	R3		4	24				
B4	R4		5	25				
B5	R5		6	26				
B6	R6		7	27				
B7	R7		8	28				
B8	R8		9	29				
B9	R9		11	32				
B10	R10		12	31				
B11	R11		13	33				
B12	R12		14	34				
B13	R13		15	35				
B14	R14		16	36				
B15	R15		17	39				
B16	R16		18	38				
B17	R17		19	37				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:42Z
Check the source to verify that this is the current version before use.



☆U.S. GOVERNMENT PRINTING OFFICE: 1982-505-022/4364

FIGURE 4. Resistance test circuit for table III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:42Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL

FIRST CLASS PERMIT NO. 12503 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Naval Electronic Systems Command
Attn: ELEX 8111
Department of the Navy
Washington, D.C. 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:42Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.
NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-M-28787/318A(EC) Modules, Standard Electronic Resistor, Analog, Key Code GRB

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ *HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?* ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426 EDITION OF 1 JAN 72 MAY BE USED UNTIL EXHAUSTED

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:42Z
Check the source to verify that this is the current version before use.